Notes on Olympiad Geometry

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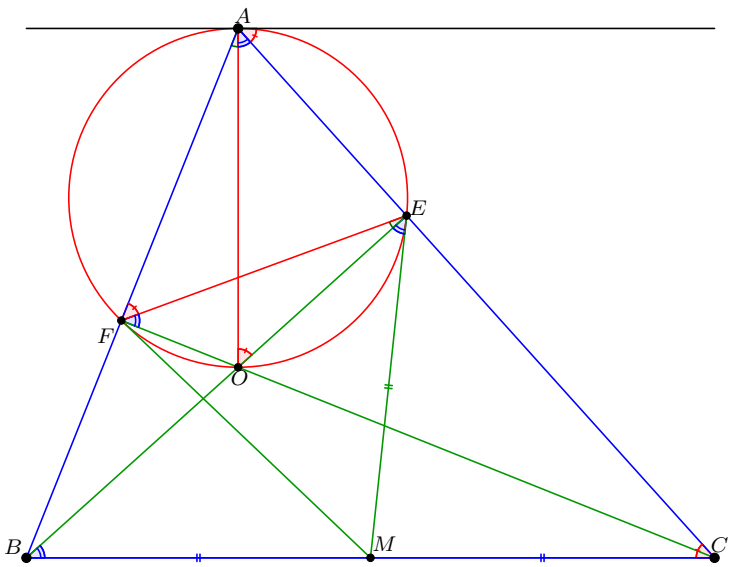
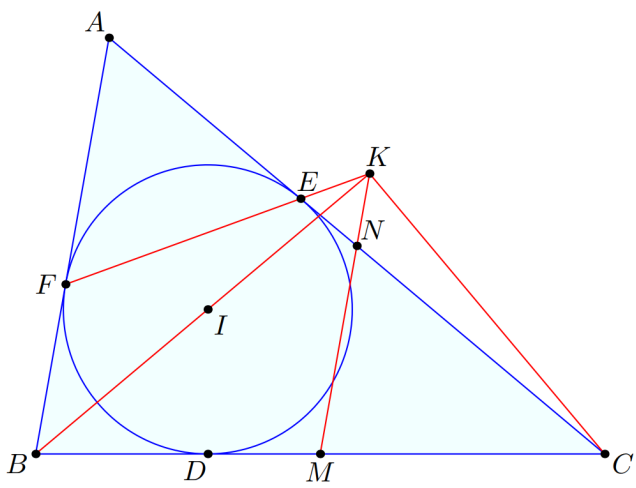
Lemma 0.1: Three Tangents Lemma

Let ABC be an acute triangle. Let BE and CF be altitudes of $\triangle ABC$, and denote by M the midpoint of BC. Prove that ME, MF, and the line through A parallel to BC are all tangents to (AEF).



Lemma 0.2: Iran Lemma

The incircle of $\triangle ABC$ is tangent to BC, CA, AB at D, E, F, respectively. Let M and N be the midpoints of BC and AC, respectively. Ray BI meets line EF at K. Show that $BK \perp CK$. Then show K lies on line MN.



Solution.

Claim — $BK \perp CK$

Proof. Since F and D are reflected over BI, the following is true.

$$\measuredangle KDC = \measuredangle BDK = \measuredangle KFB = \measuredangle EFA = \measuredangle AEF = \measuredangle CEK. \qquad \square$$

Claim — K lies on line MN.

Proof. Since M is the circumcenter of (BKC), $\measuredangle CMK = 2\measuredangle CBK = \measuredangle CBA$. Hence, $KM \parallel AB$. However, $MN \parallel AB$. Thus, $N \in KM$. $\square$

0.0.1 USAJMO 2011 P5

Problem Statement

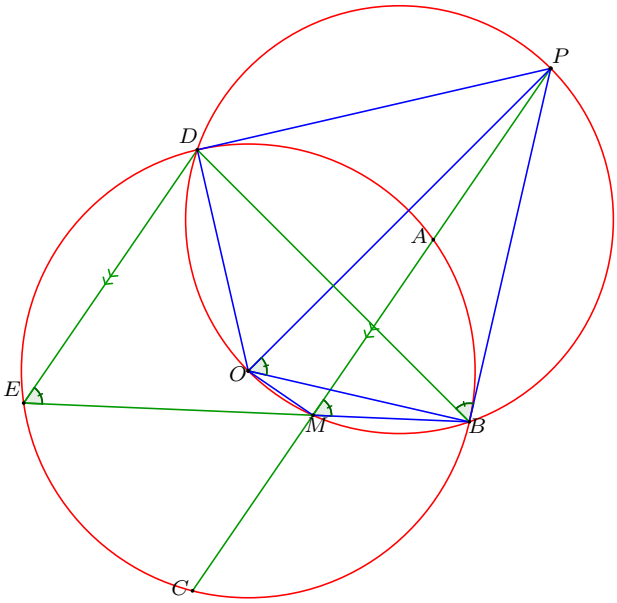
Points A, B, C, D, E lie on a circle ω and point P lies outside the circle. The given points are such that

(i) lines PB and PD are tangent to ω,

(ii) P, A, C are collinear, and

(iii) $DE \parallel AC$.

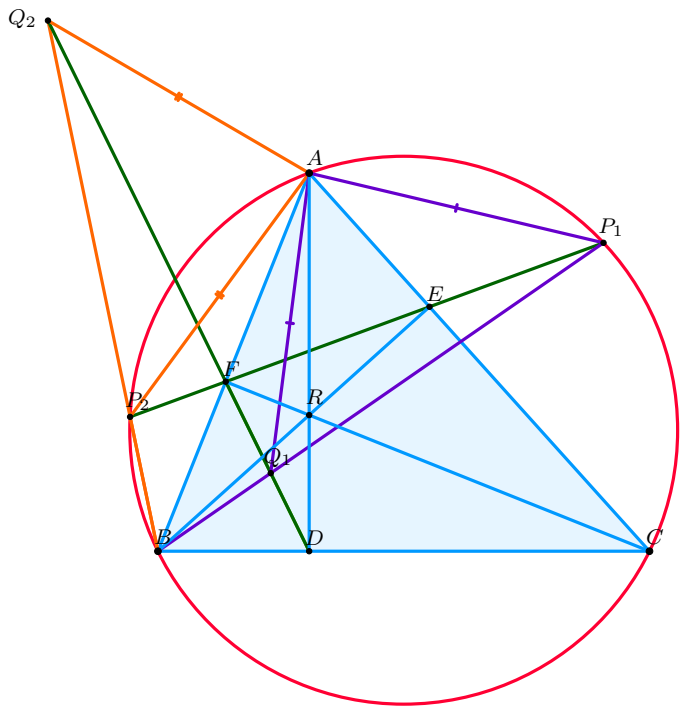
Prove that BE bisects AC.



0.0.2 Shortlist 2010 G1

Problem Statement

Let ABC be an acute triangle with D, E, F the feet of the altitudes lying on BC, CA, AB respectively. One of the intersection points of the line EF and the circumcircle is P. The lines BP and DF meet at point Q. Prove that $AP = AQ$.



Let $\measuredangle$ denote directed angles mod 180°. The line EF meets the circumcircle at two points. Directed angles allow us to treat both at once, so we fix one of them.

Our goal is to show $\measuredangle PQA = \measuredangle APQ$, i.e., that $\triangle APQ$ is isosceles. We will first prove that A, F, P, Q are concyclic.

Since $\triangle DEF$ is an orthic triangle, $\measuredangle CFA = \measuredangle ADC = 90°$. Therefore, $FACD$ is cyclic, culminating in $\measuredangle ACD = \measuredangle AFD = \measuredangle AFQ$. However, $APCB$ is cyclic, so $\measuredangle APB = \measuredangle ACB = \measuredangle ACD$. Thus, $\measuredangle APQ = \measuredangle AFQ$. Which means that $AFPQ$ is a cyclic quadrilateral.

$FECB$ is a cyclic quadrilateral either, by the same $FACD$'s reason. Therefore, putting together everything we've seen so far:

$$\measuredangle PQA = \measuredangle PFA = \measuredangle EFB = \measuredangle ECB = \measuredangle ACB = \measuredangle APB.$$

Hence, $AP = AQ$.

0.0.3 Euler's Lemma

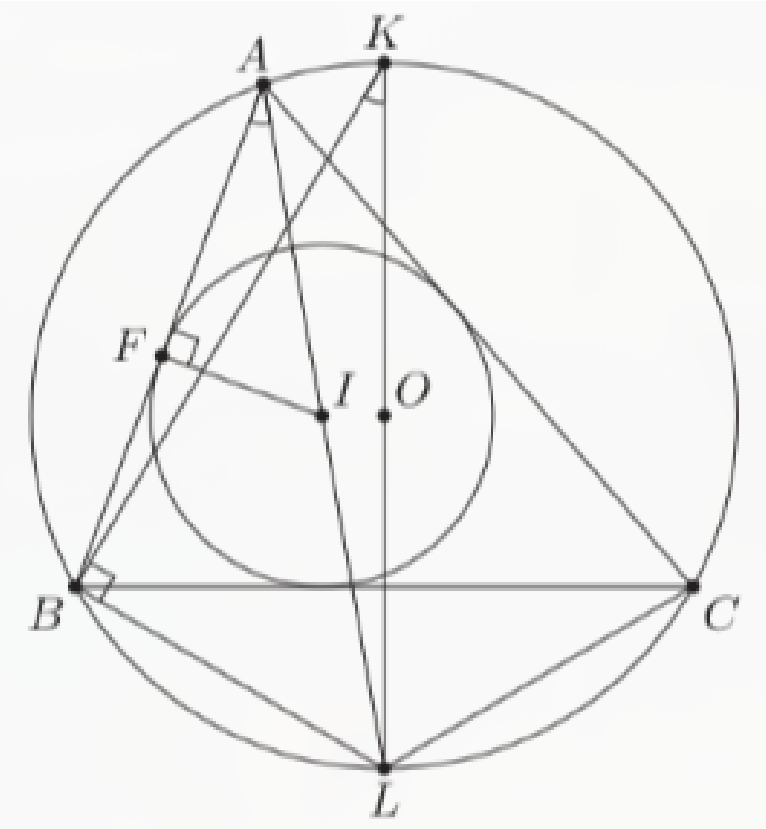


Figure 2.7C. Proving Euler's theorem.

Solution to Lemma 2.22. Let ray AI meet the circumcircle again at L and let K be the point diametrically opposite L. Let F be the foot from I to $\overline{AB}$. Notice that $\angle FAI = \angle BAL = \angle BKL$ and $\angle AFI = \angle KBL = 90^\circ$, so

$$\frac{AI}{r} = \frac{AI}{IF} = \frac{KL}{LB} = \frac{2R}{LI}$$

and hence $AI \cdot IL = 2Rr$. Because I lies inside $\triangle ABC$, we deduce the power of I with respect to (ABC) is $2Rr = R^2 - OI^2$. Consequently, $OI^2 = R(R - 2r)$. $\square$

0.0.4 All-Russian MO 2011

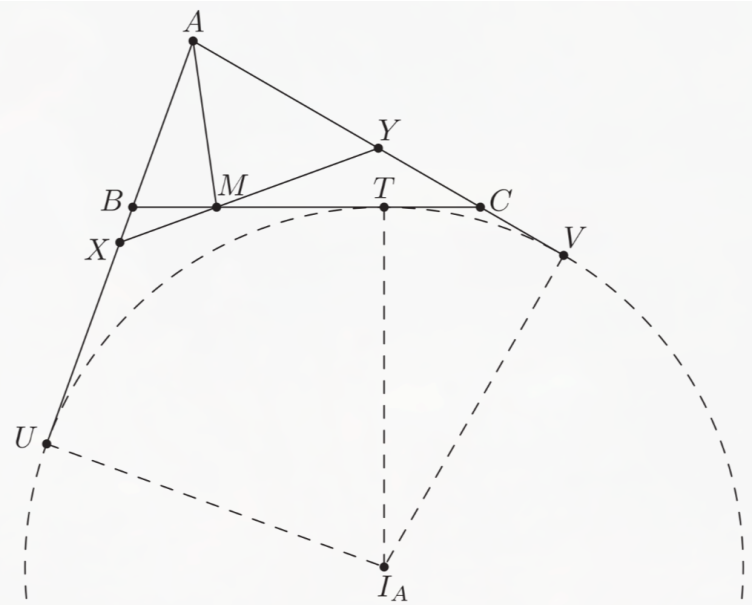


Figure 2.7E. Adding an excircle to handle the conditions.

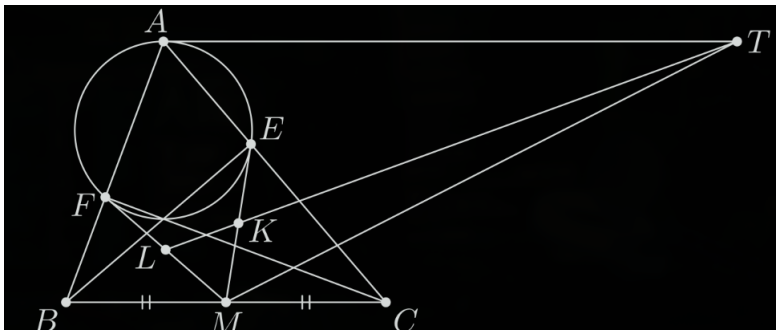
Solution to Example 2.23. Let I_A be the center of the A-excircle, tangent to $\overline{BC}$ at T, and to the extensions of $\overline{AB}$ and $\overline{AC}$ at U and V. We see that $AU = AV = s = 2$. Then $\overline{XY}$ is the radical axis of the A-excircle and the circle of radius zero at A. Therefore $AM = MT$.

Assume without loss of generality that T lies on $\overline{MC}$, as opposed to $\overline{MB}$. Then $AB + BM + MA = AB + BM + MT = AB + BT = AB + BU = AU = 2$ as desired. $\square$

0.0.5 Iran TST 2011/1 02/13/2026

Problem Statement

In acute triangle ABC, $\angle B > \angle C$. Let M be the midpoint of $\overline{BC}$ and let E and F be the feet of the altitudes from B and C, respectively. Let K and L be the midpoints of $\overline{ME}$ and $\overline{MF}$, respectively, and let T lie on line KL such that $\overline{TA} \parallel \overline{BC}$. Prove that $TA = TM$.



Let $\Gamma_1 = (AFE)$ and Γ_2 be the circumference of radius 0 centered at M. In order to prove $TA = TM$, it is enough to demonstrate that $\text{Pow}_{\Gamma_1}(T) = \text{Pow}_{\Gamma_2}(T)$.

Claim — $\overline{KL}$ is the radical axis of Γ_1 and Γ_2.

Proof. By the Three Tangents Lemma, $\overline{ME}$ and $\overline{MF}$ are both tangents to Γ_1. Since, $MK = KE$,

$$\text{Pow}_{\Gamma_1}(K) = \text{Pow}_{\Gamma_2}(K).$$

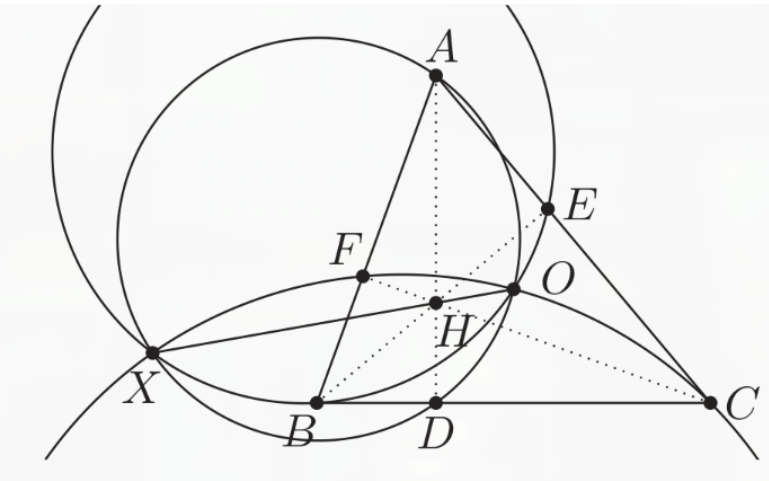
Hence, K lies inside the radical axis of Γ_1 and Γ_2. However, we can use an analogous argument to prove that L does either. Hence $\overline{KL}$ is the radical axis of Γ_1 and Γ_2. $\square$

Since K, L and M are collinear, M lies inside this radical axis too. Hence, $TA = TM$

0.0.6 02/13/26

Problem Statement

Let $\overline{AD}$, $\overline{BE}$, $\overline{CF}$ be the altitudes of a scalene triangle ABC with circumcenter O. Prove that (AOD), (BOE), and (COF) intersect at point X other than O.



Let H be the orthocenter $\triangle ABC$. Hence,

$$HA \cdot HD = HB \cdot BE = HC \cdot HF,$$

is true by considering the circumcenters of diameters AB, BC and CA. Which means that both O and H have the same power of point of all the circumferences (AOD), (BOE), and (COF).

Hence, those three circumferences have the same radical axis, resulting in another point X where they three intersect.